

Gaussian gain–loss response at a class-A dynamical domain wall

Internal campaign summary¹

¹ *Class-A fermionic adaptive-circuit project*
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This note records deterministic CPU campaign `df092e76037c014f` using the notation of Bhuiyan, Pan, and Jian [1]. The computed generator is exactly the Gaussian sector $\mathcal{L}_G = \mathcal{L}_{\text{gain}} + \mathcal{L}_{\text{loss}}$ of their effective cycle Lindbladian; it omits $\mathcal{L}_{\text{dephas.}}$ and contains neither Born trajectories nor schedule sampling. We derive the closed equation for the two-point correlation matrix G , its stationary Sylvester solution, a finite completely-positive Strang channel, and the exact retarded response to a local density-phase impulse. The data show wall-localized natural-occupation flow through half occupation and oppositely oriented signed density responses at the two walls, but also a finite mean damping gap and exponential response extinction. The calculation therefore establishes neither a gapless full Lindbladian nor a trajectory Lyapunov boundary mode of the projective adaptive circuit.

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I. SCOPE AND BOTTOM LINE

Four spectral objects must be kept distinct. First, the auxiliary two-band class-A Hamiltonian is used only to synthesize an overcomplete Wannier (OW) basis. Second, the eigenvalues $\nu \in [0, 1]$ of the stationary correlation matrix G_{ss} are natural occupations; their crossing at $\nu = 1/2$ diagnoses a closing of the stationary occupation or purity gap. Third, the positive one-particle matrix

D controls deterministic mean relaxation. Fourth, the original adaptive measurement-only circuit is a random product of record-conditioned instruments and has a distinct tangent/Lyapunov spectrum. The present campaign computes only the second and third objects for $\mathcal{L}_G$, together with a finite completely-positive discretization of that Gaussian generator.

The central result is correspondingly two-sided. At $(\alpha_{\text{run},\text{in}}, \alpha_{\text{run},\text{out}}) = (1, 30)$, two wall-localized occupation branches cross $\nu = 1/2$ with opposite k_y slopes and the bulk occupation gap remains approximately $1/2$. A local phase impulse produces signed density dipoles whose positive lobes drift in opposite y directions on the two walls. Nevertheless the smallest one-particle damping rate is $d_{\text{min}} = 0.2795656$ in the full-frame canonical case; the impulse response decays exponentially. Stationary spectral flow and gapless dynamics are therefore not interchangeable statements.

II. OW BATH AND EXACT GAUSSIAN CLOSURE

A. Auxiliary band geometry and frame operators

Following Ref. [1], let $\hat{c}(\mathbf{k}) = (\hat{c}_1(\mathbf{k}), \hat{c}_2(\mathbf{k}))^\top$. The campaign synthesizes OW states from the auxiliary parent Hamiltonian

$$\hat{H}_{\text{CI}}(\alpha_{\text{run}}) = \sum_{\mathbf{k}} \hat{c}^\dagger(\mathbf{k}) h_{\alpha_{\text{run}}}(\mathbf{k}) \hat{c}(\mathbf{k}) \quad (1)$$

and its flattened single-particle matrix

$$\begin{aligned} h_{\alpha_{\text{run}}}(\mathbf{k}) &= \hat{\mathbf{n}}_{\alpha_{\text{run}}}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \\ \mathbf{n}_{\alpha_{\text{run}}}(\mathbf{k}) &= (\sin k_x, \sin k_y, \alpha_{\text{run}} - \cos k_x - \cos k_y), \end{aligned} \quad (2)$$

with $P_{\pm}(\mathbf{k}) = (\mathbf{1} \pm h_{\alpha_{\text{run}}}(\mathbf{k}))/2$. This run convention is sign-reversed relative to the published parent Hamiltonian: throughout this note

$$\alpha_{\text{PRR}} = -\alpha_{\text{run}}. \quad (3)$$

The archived labels and figures retain $\alpha_{\text{run}} = 1, 30, \dots$. For each center $\mathbf{r} = (x, y)$, the code uses the *uniform*

parent at its local $\alpha_{\text{run}}(x)$; it never diagonalizes a spatial domain-wall Hamiltonian. It projects the trial orbitals $|\tau_A\rangle = (1, 1)^T/\sqrt{2}$ and $|\tau_B\rangle = (1, -1)^T/\sqrt{2}$ onto $P_{\pm}$ and Fourier transforms them to $|W_{\mathbf{r},\nu,\pm}\rangle$, $\nu = A, B$.

Let $\mathbf{r}'$ denote a support cell. With

$$\begin{aligned} M_{n_{\text{shell}}}(\mathbf{r}' - \mathbf{r}) &= \mathbf{1}\{|\Delta x|_{\text{per}}, |\Delta y|_{\text{per}} \leq n_{\text{shell}}\}, \\ M_{\text{DW}}(r'_x; r_x) &= \mathbf{1}\{r'_x, r_x \text{ lie in the same domain}\}, \end{aligned} \quad (4)$$

the normalized mode actually used is

$$|\widetilde{W}_{\mathbf{r},\nu,\pm}\rangle = \frac{M_{n_{\text{shell}}} M_{\text{DW}} |W_{\mathbf{r},\nu,\pm}\rangle}{\|M_{n_{\text{shell}}} M_{\text{DW}} |W_{\mathbf{r},\nu,\pm}\rangle\|}. \quad (5)$$

Here $M_{n_{\text{shell}}} = 1$ denotes the full frame, and $M_{\text{DW}} = 1$ when hard wall truncation is disabled. The OW states remain nonorthogonal and overcomplete. Define their positive frame operators

$$F_{\pm} = \sum_{\mathbf{r}, \nu=A,B} |\widetilde{W}_{\mathbf{r},\nu,\pm}\rangle \langle \widetilde{W}_{\mathbf{r},\nu,\pm}|, \quad (6)$$

which are not projectors in general.

The canonical profile has $N_x = 20$, walls at $x = 5, 15$, and $\alpha_{\text{run,out}} = 30$. All coefficients are periodic and uniform along y . Since the shell mask depends only on relative periodic displacement and the wall mask only on x ,

$$\begin{aligned} T_y |\widetilde{W}_{x,y,\nu,\pm}\rangle &= |\widetilde{W}_{x,y+1,\nu,\pm}\rangle, \\ [T_y, F_{\pm}] &= 0, \quad [T_y, D] = [T_y, S] = 0. \end{aligned} \quad (7)$$

This is an exact symmetry of the deterministic generator. A fixed adaptive trajectory or fixed noncommuting raster order has no corresponding guarantee.

Using $\hat{c}_{x\sigma}(k_y) = N_y^{-1/2} \sum_y e^{-ik_y y} \hat{c}_{xy\sigma}$, translation symmetry reduces every single-particle matrix to $2N_x = 40$ dimensional blocks. In the published correlation convention,

$$G_{x\sigma, x'\sigma'}(k_y) = \frac{1}{N_y} \sum_{y, y'} e^{ik_y(y-y')} G_{xy\sigma, x'y'\sigma'}. \quad (8)$$

B. Gaussian sector of the effective Lindbladian

The OW fermion modes and their number stabilizers are

$$\begin{aligned} \hat{\chi}_{\mathbf{r},\nu,\pm} &= \sum_{\mathbf{r}', \sigma} \widetilde{W}_{\mathbf{r},\nu,\pm}(\mathbf{r}', \sigma) \hat{c}_{\mathbf{r}', \sigma}, \\ \hat{N}_{\mathbf{r},\nu,\pm} &= \hat{\chi}_{\mathbf{r},\nu,\pm}^\dagger \hat{\chi}_{\mathbf{r},\nu,\pm}. \end{aligned} \quad (9)$$

Distinct OW stabilizers generally do not commute. With

$$\mathcal{D}[\hat{L}](\hat{\rho}) = \hat{L} \hat{\rho} \hat{L}^\dagger - \frac{1}{2} \{\hat{L}^\dagger \hat{L}, \hat{\rho}\}, \quad (10)$$

the campaign generator is

$$\begin{aligned} \mathcal{L}_{\text{gain}} &= \bar{n}_a \sum_{\mathbf{r}, \nu} \mathcal{D}[\hat{\chi}_{\mathbf{r},\nu,-}^\dagger], \\ \mathcal{L}_{\text{loss}} &= (1 - \bar{n}_a) \sum_{\mathbf{r}, \nu} \mathcal{D}[\hat{\chi}_{\mathbf{r},\nu,+}], \\ \mathcal{L}_G &= \mathcal{L}_{\text{gain}} + \mathcal{L}_{\text{loss}}. \end{aligned} \quad (11)$$

It is the Gaussian part of the paper's effective generator

$$\mathcal{L}_{\text{cycle}} = \mathcal{L}_G + \mathcal{L}_{\text{dephas.}} \quad (12)$$

For reference, the omitted sector is

$$\begin{aligned} \mathcal{L}_{\text{dephas.}} &= (2 - \bar{n}_a) \sum_{\mathbf{r}, \nu} \mathcal{D}[\hat{N}_{\mathbf{r},\nu,-}] \\ &\quad + (1 + \bar{n}_a) \sum_{\mathbf{r}, \nu} \mathcal{D}[\hat{N}_{\mathbf{r},\nu,+}]. \end{aligned} \quad (13)$$

That non-Gaussian many-body sector was not simulated here; neither was the ordered projective adaptive circuit.

Fix the paper's two-point convention

$$G_{ij} = \text{Tr}(\hat{\rho} \hat{c}_i^\dagger \hat{c}_j). \quad (14)$$

The production solver stores the transposed convention, so the exact bridge is

$$C_{\text{arch}} = G^\top = G^*. \quad (15)$$

The last equality uses Hermiticity. Consequently, transposition changes eigenvectors by complex conjugation but leaves natural occupations and spatial weights invariant.

For $\hat{G}_{ij} = \hat{c}_i^\dagger \hat{c}_j$, the adjoint identity

$$\mathcal{D}^\dagger[\hat{L}](\hat{O}) = \frac{1}{2} (\hat{L}^\dagger [\hat{O}, \hat{L}] + [\hat{L}^\dagger, \hat{O}] \hat{L}) \quad (16)$$

and the canonical anticommutation relations close the dynamics exactly on G . Because of Eq. (14), the OW frame enters complex conjugated:

$$\dot{G} = -DG - GD + S, \quad (17)$$

$$\begin{aligned} D &= \frac{1}{2} [\bar{n}_a F_-^* + (1 - \bar{n}_a) F_+^*], \\ S &= \bar{n}_a F_-^*. \end{aligned} \quad (18)$$

For one mode this gives $\dot{g} = \gamma^+(1 - g) - \gamma^- g$, fixing all factors and signs.

C. Stationary solution, damping, and occupation topology

The stationary correlation matrix solves the Sylvester equation

$$DG_{\text{ss}} + G_{\text{ss}}D = S. \quad (19)$$

Writing $D = U \text{diag}(d_a) U^\dagger$ gives the explicit solution

$$(U^\dagger G_{\text{ss}} U)_{ab} = \frac{(U^\dagger S U)_{ab}}{d_a + d_b}. \quad (20)$$

Translation along y reduces this to independent $2N_x = 40$ dimensional blocks $D(k_y)$ and $G(k_y)$. A two-point perturbation obeys

$$\delta G(t) = e^{-Dt} \delta G(0) e^{-Dt}, \quad (21)$$

so its elementary decay rates are sums $d_a + d_b$. The one-particle $d_{\min}$ is therefore a stability proxy, while the correlation-sector gap begins at $2d_{\min}$.

By contrast, the natural occupations $\nu_j(k_y)$ are eigenvalues of $G(k_y)$. When the bulk is separated from $\nu = 1/2$, one may flatten $Q_G = \mathbf{1} - 2G_{ss}$ or introduce the Gaussian modular matrix

$$h_{\text{mod}} = \log[(\mathbf{1} - G_{ss})G_{ss}^{-1}]. \quad (22)$$

A branch crossing $\nu = 1/2$ is a zero crossing of either classifying representative. It says nothing by itself about a zero of D or of a trajectory Lyapunov exponent [6, 7].

III. FINITE CHANNEL AND CONTINUOUS LIMIT

The finite map is not the projective adaptive circuit. It is a completely-positive Gaussian channel that discretizes Eqs. (11) and (17). Define $\Gamma_{\text{gain}} = \bar{n}_a F_-^*$ and $\Gamma_{\text{loss}} = (1 - \bar{n}_a) F_+^*$. A loss-only interval acts as

$$\mathcal{E}_{\text{loss}}(t) : G \mapsto A_- G A_-^\dagger, \quad A_- = e^{-\Gamma_{\text{loss}} t/2}, \quad (23)$$

while a gain-only interval acts as

$$\mathcal{E}_{\text{gain}}(t) : G \mapsto A_+ G A_+^\dagger + \mathbf{1} - A_+ A_+^\dagger, \quad A_+ = e^{-\Gamma_{\text{gain}} t/2}. \quad (24)$$

One step of size p is the ordered Strang composition

$$\mathcal{E}_p = \mathcal{E}_{\text{gain}}(p/2) \circ \mathcal{E}_{\text{loss}}(p) \circ \mathcal{E}_{\text{gain}}(p/2). \quad (25)$$

Composition acts right to left. Writing $R_p = e^{-\Gamma_{\text{gain}} p/4}$ and $L_p = e^{-\Gamma_{\text{loss}} p/2}$, the complete affine step is

$$\begin{aligned} \mathcal{E}_p(G) &= A_p G A_p^\dagger + B_p, \quad A_p = R_p L_p R_p, \\ B_p &= R_p L_p (\mathbf{1} - R_p^2) L_p R_p + \mathbf{1} - R_p^2. \end{aligned} \quad (26)$$

Exact affine binary exponentiation evaluates $\mathcal{E}_p^{t/p}$ without a long time loop. At fixed physical time, Strang splitting has global error $O(p^2)$. This controlled comparison is why the production bundle is called *hybrid*: the continuous generator supplies the stationary state and main response, while the finite channel verifies convergence to it.

IV. EXACT RESPONSE TO A LOCAL DENSITY-PHASE IMPULSE

A. Origin of δG

Let P_s project onto the two physical orbitals at source cell $s = (x_w, y_0)$ and let $\hat{n}_s = \hat{c}^\dagger P_s \hat{c}$. The many-body

impulse

$$\hat{U}_\epsilon = e^{-i\epsilon \hat{n}_s} \quad (27)$$

induces the one-particle rotation $u_\epsilon = e^{-i\epsilon P_s}$ because $[\hat{n}_s, \hat{c}] = -P_s \hat{c}$. In the correlation convention (14), the kicked state is therefore

$$G_\epsilon(0) = u_\epsilon^\dagger G_{ss} u_\epsilon. \quad (28)$$

The symbol δG is simply the tangent to this one-parameter family:

$$G_\epsilon(t) = G_{ss} + \epsilon \delta G(t) + O(\epsilon^2), \quad (29)$$

$$\delta G(0) = \partial_\epsilon G_\epsilon(0)|_0 = +i[P_s, G_{ss}]. \quad (30)$$

It is not an extra stochastic correlator or an assumed source. Since the kick changes the initial state but not D or S , subtracting the stationary equation from Eq. (17) cancels the affine source:

$$\delta \dot{G} = -D\delta G - \delta G D. \quad (31)$$

Equations (21) and (30) then give

$$\boxed{\delta G(t) = +i e^{-Dt} [P_s, G_{ss}] e^{-Dt}}. \quad (32)$$

For a density probe P_r ,

$$\boxed{\chi_{rs}(t) = +i\theta(t) \text{tr}[P_r e^{-Dt} [P_s, G_{ss}] e^{-Dt}]}. \quad (33)$$

Because local density projectors commute, $\chi_{rs}(0) = +i \text{tr}([P_r, P_s] G_{ss}) = 0$: the kick initially rotates coherences, and the dissipative propagator subsequently converts them into density.

The finite central difference is also analytic. Since $P_s^2 = P_s$,

$$\frac{G_{+\epsilon}(0) - G_{-\epsilon}(0)}{2\epsilon} = +i \frac{\sin \epsilon}{\epsilon} [P_s, G_{ss}]. \quad (34)$$

Thus $\epsilon = 10^{-3}$ merely multiplies the derivative by $\text{sinc } \epsilon = 1 - O(10^{-7})$; no pair of numerical trajectory simulations is required.

B. Spectral, frequency, and momentum representations

In the D eigenbasis, Eq. (33) becomes

$$\chi_{rs}(t) = +i\theta(t) \sum_{a,b} e^{-(d_a+d_b)t} \langle b|P_r|a\rangle \langle a|[P_s, G_{ss}]|b\rangle. \quad (35)$$

With Fourier convention $\int_0^\infty dt e^{i\omega t}$,

$$\chi_{rs}(\omega) = +i \sum_{a,b} \frac{\langle b|P_r|a\rangle \langle a|[P_s, G_{ss}]|b\rangle}{d_a + d_b - i\omega}. \quad (36)$$

All mean-response poles are therefore at $\omega = -i(d_a + d_b)$. A half-occupation crossing of G_{ss} does not create a zero-frequency pole when $d_{\min} > 0$.

For a source at y_0 and a wall-window probe at y , let p_s, p_w be their x -orbital projectors and $r = y - y_0$. Exact y -translation symmetry gives

$$\chi_w(r, t) = +\frac{i}{N_y^2} \sum_{k, q} e^{i(k-q)r} \times \text{tr} \left[p_w e^{-D(k)t} (p_s G(q) - G(k) p_s) e^{-D(q)t} \right]. \quad (37)$$

The heatmap is a direct finite evaluation of Eq. (37). There is no continuous-time integration.

C. Rank-four implementation

Writing $P_s = EE^\dagger$ with two source-orbital columns and $A(t) = e^{-Dt}$, define $X = AE$ and $Y = AG_{ss}E$. Then

$$\begin{aligned} \delta G(t) &= +i(XY^\dagger - YX^\dagger), \\ \delta G_{ii}(t) &= +2 \text{Im} \sum_{\sigma=1}^2 Y_{i\sigma} X_{i\sigma}^*. \end{aligned} \quad (38)$$

After applying Eq. (15), the code propagates only E and $G_{ss}E$ through momentum blocks and uses Eq. (38); it never materializes a kicked correlation history. In particular, $\delta G_{ii} = \delta(C_{\text{arch}})_{ii}$, so the plotted density response is unchanged by the convention migration. This is an exact Gaussian Kubo impulse response [3, 4], not Monte Carlo sampling.

V. CAMPAIGN DESIGN AND PROVENANCE

The manifest records complete SHA-256 hashes for the solver and configuration; their identifying prefixes are `3efc79ee1434` and `aa2956995cac`. Momentum-resolved arrays were archived by policy only for the full frame. Figure 1 therefore reconstructs the shell-one block from the pinned solver without altering the production archive.

VI. RESULTS

A. Wall occupation spectral flow

For every saved momentum block the scatter points in Fig. 1 are defined by

$$G_{ss}(k_y) |\phi_j(k_y)\rangle = \nu_j(k_y) |\phi_j(k_y)\rangle. \quad (39)$$

Let $\mathcal{W}_a = \{x_a - 1, x_a, x_a + 1\}$ be the three-column window around wall a . Its mode weight is

$$W_j^{(a)}(k) = \sum_{x \in \mathcal{W}_a, \sigma} |\phi_{j, x\sigma}(k)|^2. \quad (40)$$

Among modes with $W_j^{(a)} \geq 0.25$, the colored branch is selected separately at each k by

$$j_a(k) = \arg \min_j \left\{ |\nu_j(k) - \tfrac{1}{2}| + 0.02[1 - W_j^{(a)}(k)] \right\}. \quad (41)$$

If no mode passes the threshold, the maximum-weight mode is used. The displayed slope s_a is the ordinary least-squares coefficient in $\nu_{j_a}(k) = s_a k + b_a$ for

$$|k| \leq k_c, \quad k_c = \max \left(2.5 \frac{2\pi}{N_y}, 0.22 \right). \quad (42)$$

This selector is not global overlap tracking and is not itself a topological index. At the exact $k = 0$ degeneracy the eigensolver may rotate the two wall states into each other.

For the canonical full-frame $N_y = 64$ case,

$$\begin{aligned} \min_{k, j} |\nu_j(k) - \tfrac{1}{2}| &= 2.22 \times 10^{-16}, \\ \Delta_{\text{bulk}} &= 0.49999999993. \end{aligned} \quad (43)$$

The selected slopes are $\partial_{k_y} \nu_w = \pm 0.949605$, and the stationary Gaussian entropy is $S_G/N_y = 0.317152$. The shell-one result remains sharply distinguished: its half gap is 3.33×10^{-10} , bulk gap 0.496999, and selected wall slopes ± 0.659914 . Shell two is intermediate, with slopes ± 0.823598 . Thus finite n_{shell} has little visible effect on the many eigenvalues already pinned exponentially close to 0 or 1, but it systematically reduces the wall-branch slope and broadens the crossing in momentum. Uniform topological and uniform trivial controls have half gaps 0.4969–0.5 and no wall response.

The four alpha-scan panels use the following estimators. With $\mathcal{W} = \mathcal{W}_1 \cup \mathcal{W}_2$ and $W_j^{\mathcal{W}} = \sum_{x \in \mathcal{W}, \sigma} |\phi_{j, x\sigma}|^2$,

$$\begin{aligned} \Delta_{1/2} &= \min_{k, j} |\nu_j(k) - \tfrac{1}{2}|, \\ \Delta_{\text{bulk}} &= \min_{k, j: W_j^{\mathcal{W}}(k) < 0.25} |\nu_j(k) - \tfrac{1}{2}|, \end{aligned} \quad (44)$$

$$\frac{S_G}{N_y} = -\frac{1}{N_y} \sum_{k, j} [\nu_j \ln \nu_j + (1 - \nu_j) \ln(1 - \nu_j)]. \quad (45)$$

Thus “bulk” is a wall-weight exclusion within the same domain-wall state, not a separate uniform calculation, and S_G/N_y is entropy per circumference, not per site. For the localization panel let $\mathcal{M}$ be the $2N_y$ pairs (k, j) closest to half occupation and define

$$p_x = \frac{1}{|\mathcal{M}|} \sum_{(k, j) \in \mathcal{M}, \sigma} |\phi_{j, x\sigma}(k)|^2, \quad \mathcal{L}_{\text{mid}} = \frac{\sum_{x \in \mathcal{W}} p_x}{\sum_x p_x}. \quad (46)$$

Finally, if $D(k)|u_a(k)\rangle = d_a(k)|u_a(k)\rangle$, a correlation eigen-operator decays at

$$\gamma_{ab}(k) = d_a(k) + d_b(k), \quad \gamma_{\min} = \min_{\gamma_{ab} > 10^{-12}} \gamma_{ab}. \quad (47)$$

The lower-right panel plots $\gamma_{\min}$; at the canonical endpoint it is $2d_{\min} = 0.5591313$, not a Hamiltonian excitation gap.

TABLE I. Production design. Full frame denotes $n_{\text{shell}} = \text{None}$. All α values are α_{run} from Eq. (3). The static sweep keeps a geometric α_{run} interface; for $\alpha_{\text{run},\text{in}} > 2$ both regions are topologically trivial even though the spatial step remains.

Run identity	ID df092e76037c014f; 66/66 cases complete; complex128 CPU; 48 workers; one BLAS thread each.
Main geometry	$N_x = 20$; periodic y ; canonical walls $x = 5, 15$; $\alpha_{\text{run},\text{out}} = 30$; $\bar{n}_a = 1/2$; maximally mixed initialization.
Static scan	$N_y = 64$; $\alpha_{\text{run},\text{in}} = 1, 1.5, 1.75, 1.875, 2.125, 2.25, 2.5, 3$; $n_{\text{shell}} = 1, 2, \text{None}$; domain_wall=True ; dw_truncation=True .
Response endpoint	$\alpha_{\text{run},\text{in}} = 1$; $N_y = 32, 48, 64$; all three shells; truncation on and off; $\epsilon = 10^{-3}$; $t \in [0, N_y/2]$ in steps 0.25; velocity fit $2 \leq t \leq 0.375 N_y$.
Finite channel	$p = 1, 1/2, 1/4, 1/8, 1/16$ at fixed physical time; exact affine powers of the completely-positive Strang step.
Controls	Uniform topological and trivial geometries; $\bar{n}_a = 0.45, 0.55$; empty and filled initial states; legacy wall rule; separate 4×6 dephasing-closure checks.
Storage contract	Zero trajectories and schedule samples; no final correlation matrix or history; compact selected-observable NPZ plus JSON metadata and checksums.

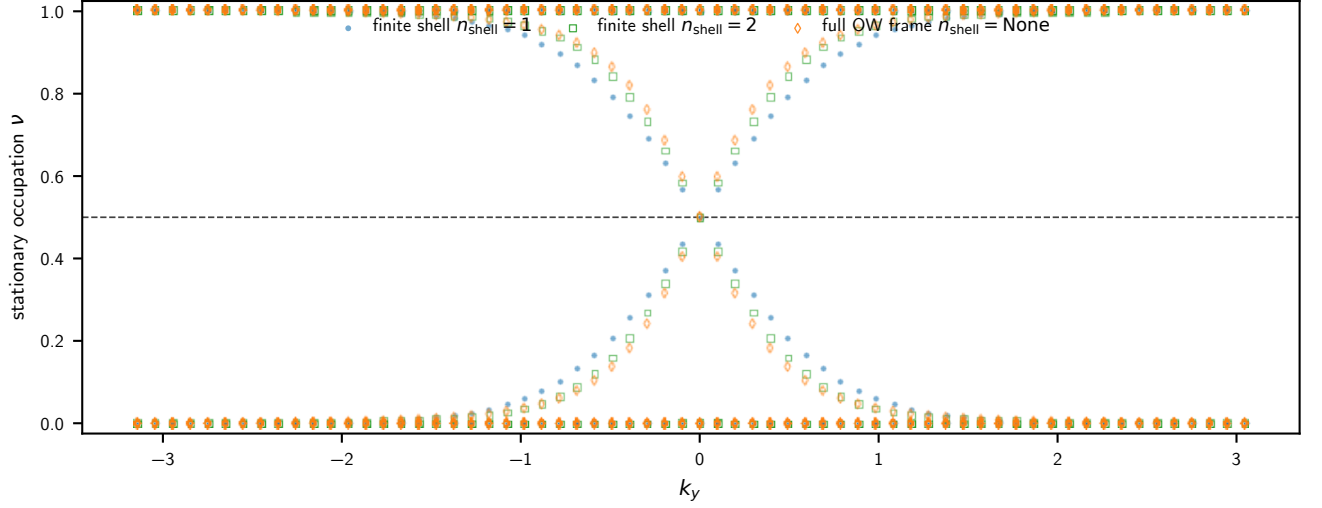


FIG. 1. Stationary k_y -resolved natural occupations at $(\alpha_{\text{run},\text{in}}, \alpha_{\text{run},\text{out}}) = (1, 30)$ with domain-wall truncation. Every eigenvalue of $G_{\text{ss}}(k_y)$ is shown without branch-selection or fit overlays: filled blue circles denote $n_{\text{shell}} = 1$, open green squares denote $n_{\text{shell}} = 2$, and open orange diamonds denote the full OW frame. These are natural occupations of the mixed steady state of $\mathcal{L}_G$. The three spectra largely coincide at $\nu = 0, 1$; their visible difference is concentrated in the wall branches crossing $\nu = 1/2$, which become successively steeper as the retained OW range increases.

Figure 2 tests whether the wall feature follows the phase mismatch rather than the mere spatial step. For shell one the half gap grows from 3.33×10^{-10} at $\alpha_{\text{run},\text{in}} = 1$ to $7.39 \times 10^{-4}, 0.0593, 0.1719$ at $1.5, 1.75, 1.875$, then to 0.3935 at 2.125 and 0.4933 at 3 . The full-frame gap changes from 2.22×10^{-16} at 1 to 0.0238 at 1.875 , 0.4802 at 2.125 , and 0.49946 at 3 . Finite shells smear the approach to the parent transition; the main qualitative change remains tied to loss of the topological mismatch.

B. What was already known about such spectra

Chiral-looking eigenvalue branches of a mixed Gaussian state are an established open-system idea, although the present OW domain-wall realization is different. For a state with correlation matrix G , the purity or modular representative $Q_G = \mathbf{1} - 2G$ has the same eigenvectors

and maps the occupation midpoint $\nu = 1/2$ to zero. Bndich and Diehl formalized topology under precisely such spectral constraints on density matrices [8]; the modular-band and purity-gap formulation was developed into an observable mixed-state geometric phase by Bardyn *et al.* [9]. Bardyn *et al.* then formulated dissipative bulk-edge correspondence in terms of two independent objects: an edge can close the steady-state purity gap, the damping gap, or both [10]. For two-dimensional class-A Gaussian Chern states, Goldstein explicitly discussed topological edge modes of the single-particle density matrix and simultaneously proved restrictions on finite-range, finite-rate preparation of pure Chern states [11].

Thus Fig. 1 has a clear precedent: it is the mixed-state analogue of a Chern edge spectrum, with $\nu - 1/2$ replacing the energy. What is *not* inherited is a Hamiltonian group velocity. Shavit and Goldstein showed in dissipatively prepared Chern states that edge-current confinement and the

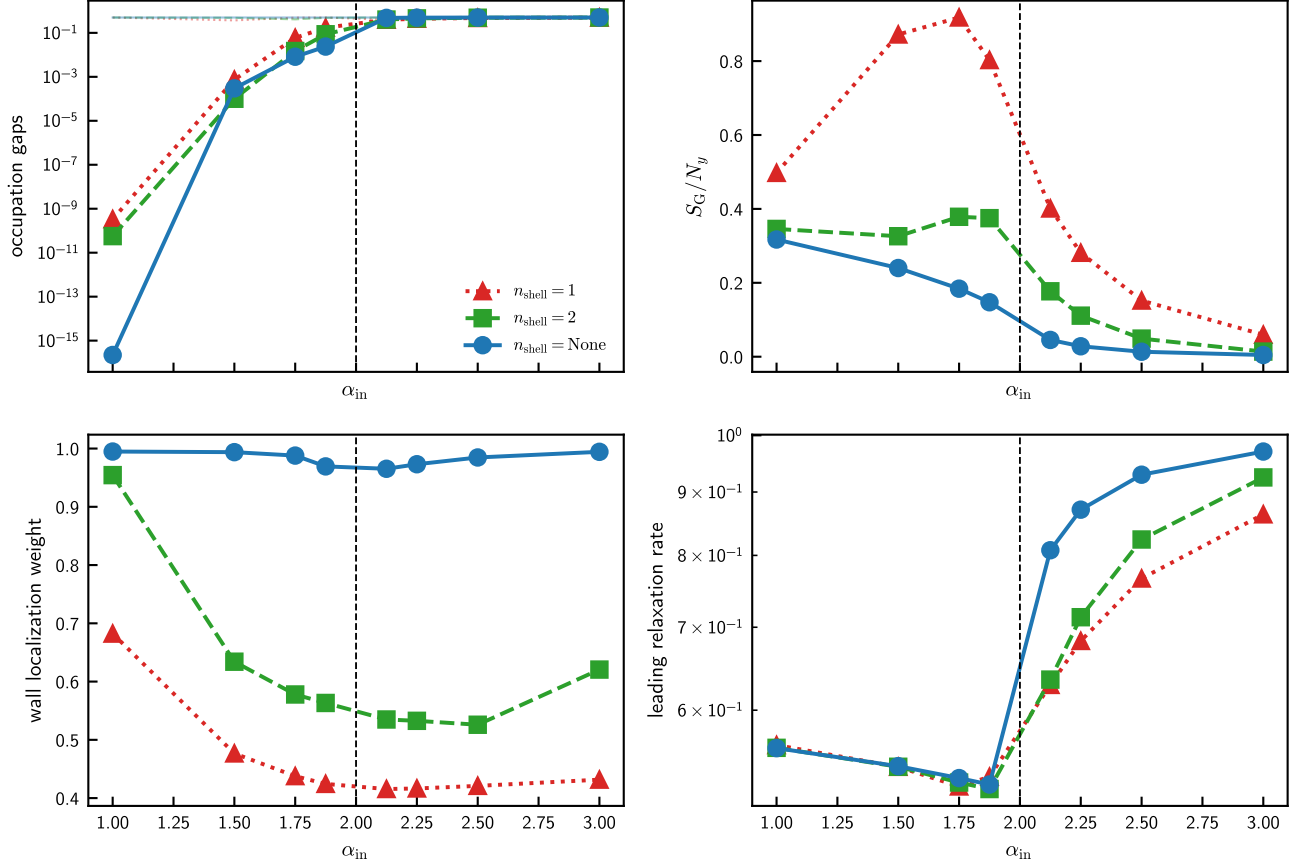


FIG. 2. Static $\alpha_{\text{run},\text{in}}$ sweep at $N_y = 64$, with $\alpha_{\text{run},\text{out}} = 30$, the geometric interface present, and domain-wall truncation on. The archived graphic abbreviates $\alpha_{\text{run},\text{in}}$ as α_{in} ; Eq. (3) gives the PRR sign convention. The dashed line marks the run-convention parent transition $\alpha_{\text{run}} = 2$. Top left: global half-occupation gap and bulk reference. Top right: stationary Gaussian entropy per circumference. Bottom left: localization weight $\mathcal{L}_{\text{mid}}$. Bottom right: smallest positive correlation-sector decay rate γ_{min} . The dashed line is only a parent-model reference: $\alpha_{\text{run}} = 2$ itself was not sampled.

relation between the density-matrix Chern number and Hall conductance can fail, and that a physical current requires additional dynamical structure [12]. Our opposite occupation slopes and signed-kick orientations are consequently suggestive state-level chirality, while the finite D gap demonstrates that they are not undamped mean-channel edge modes. Chiral topological response has been derived for driven-open Chern states under stronger purity and particle-number-conservation assumptions [13]; the gain/loss bath here satisfies neither assumption.

C. Directional but extinguished density response

Figure 3 contains two distinct impulse experiments. For panel a , both source orbitals are kicked at that panel's wall,

$$P_{a,0} = \sum_{\sigma} |x_a, y_0, \sigma\rangle \langle x_a, y_0, \sigma|, \\ \delta G_a^{(\epsilon)}(t) = +i \operatorname{sinc}(\epsilon) e^{-Dt} [P_{a,0}, G_{\text{ss}}] e^{-Dt}. \quad (48)$$

The sinc factor is the exact central-difference factor in Eq. (34). The real-space response and plotted three-column profile are

$$\chi_a(x, r, t) = \sum_{\sigma} [\delta G_a^{(\epsilon)}(t)]_{x, y_0+r, \sigma; x, y_0+r, \sigma}, \\ \chi_a^{\text{wall}}(r, t) = \sum_{x \in \mathcal{W}_a} \chi_a(x, r, t). \quad (49)$$

The full-density norm and wall retention are, respectively,

$$\mathcal{N}_a(t) = \sum_{x,r} |\chi_a(x, r, t)|, \\ \mathcal{R}_a(t) = \frac{\sum_{x \in \mathcal{W}_a, r} |\chi_a(x, r, t)|}{\mathcal{N}_a(t)}. \quad (50)$$

On the periodic ring define $\tilde{r} = ((r + N_y/2) \bmod N_y) - N_y/2$ and $[z]_+ = \max(z, 0)$. The white points in the figure are the positive-lobe center

$$y_{+,a}(t) = \frac{\sum_r \tilde{r} [\chi_a^{\text{wall}}(r, t)]_+}{\sum_r [\chi_a^{\text{wall}}(r, t)]_+}. \quad (51)$$

The yellow line is its ordinary least-squares fit over $2 \leq t \leq 0.375N_y$, retaining times for which $\mathcal{N}_a(t) > \max(10^{-8}\mathcal{N}_{a,\max}, 10^{-15})$. Its R^2 is deterministic fit quality, not a statistical confidence interval.

The response begins at numerical zero in density, grows as coherences rotate into diagonal weight, reaches an L^1 norm 0.02673 at $t = 1.25$, and then falls to 7.61×10^{-4} at $t = 8$, 4.96×10^{-6} at $t = 16$, and 3.15×10^{-10} at $t = 32$. Its final norm is 1.18×10^{-8} of the peak. During the fit the response remains essentially at its source wall (retention > 0.999), so decay is not caused by leakage across the slab.

The centers of the positive lobes fit

$$v_1 = +0.077455, \quad v_2 = -0.077455, \quad R^2 = 0.99753 \quad (52)$$

for the canonical full frame. These are velocities of a signed, decaying shape, not Hamiltonian group velocities and not quantized charge transport. Absolute right/left weight asymmetry is zero; orientation resides in the signed dipole. Because D is Hermitian positive, asymmetric spectral filtering and complex chiral matrix elements can move this signed center without producing an oscillatory real-time edge quasiparticle.

The left panel of Fig. 4 plots these raw OLS slopes against $1/N_y$; no thermodynamic extrapolation is performed. Solid curves enforce Eq. (5)'s domain-wall mask, while dashed curves allow OW support across the wall. The right panel compares finite and continuous responses at $t_f = N_y/2$ using

$$E_\chi(p) = \frac{\|\chi_p(t_f) - \chi_{\text{cont}}(t_f)\|_2}{\|\chi_{\text{cont}}(t_f)\|_2}. \quad (53)$$

Here χ_p is evaluated with the homogeneous factor $A_p^{t_f/p}$ of the affine channel, obtained by exact binary exponentiation. The separately quoted state error at $t_G = 2N_y$ is

$$E_G(p) = \frac{\|G_p(t_G) - G_{\text{cont}}(t_G)\|_F}{\|G_{\text{cont}}(t_G)\|_F}, \quad t_G = 2N_y. \quad (54)$$

At $N_y = 32, 48, 64$, canonical full-frame speeds are respectively 0.06753, 0.07499, 0.07745 in magnitude; without hard wall truncation they are 0.03222, 0.03594, 0.03803. At $N_y = 64$ with truncation, the magnitudes increase systematically from shell one to shell two to full frame: 0.05104, 0.06890, 0.07745. The orientation is robust, while its magnitude is not universal.

D. Convergence, controls, and numerical validation

For the canonical full-frame case, the finite-channel state error relative to the continuous solution decreases from 6.32×10^{-4} at $p = 1$ to 2.49×10^{-6} at $p = 1/16$; the corresponding response error decreases from 4.11×10^{-7} to 1.60×10^{-9} . Log-log fits give orders 1.99697 for E_G and 2.00088 for E_χ , verifying global $O(p^2)$ convergence.

Empty, filled, and maximally mixed initial conditions converge to the same stationary observables. Biasing the bath to $\bar{n}_a = 0.45$ or 0.55 symmetrically opens the canonical full-frame half-occupation gap to 0.04755.

All 66 manifest cases completed and all 132 case-file checksums match. Across primary Gaussian cases, the maximum stationary relative residual was 3.98×10^{-15} , the maximum frame norm error 4.88×10^{-15} , the recorded translation residual zero, the largest continuous-correlation physicality violation 6.44×10^{-15} , and the largest finite-channel physicality violation 2.83×10^{-13} . The small 4×6 dephasing cases are historical physicality controls from a different bath family and are not evidence for the primary gain/loss response.

VII. INTERPRETATION AND LIMITATIONS

The calculation is a concrete quasi-free instance of Lindbladian Kubo response [3, 4]. For an impulsive Hamiltonian perturbation $-i[\hat{n}_s, \cdot]$, Eq. (33) is the decaying-sector response of a unique stationary state. A sustained or adiabatic flux perturbation would instead require perturbing D and S and solving

$$D\delta G + \delta G D - i\omega\delta G = \delta S - \delta D G_{\text{ss}} - G_{\text{ss}} \delta D, \quad (55)$$

with a carefully defined open-system current; such flux definitions are not unique in general [5]. The local kick used here is nevertheless well-defined and its full response is given by Eqs. (33)–(37).

The campaign establishes three facts: a stationary wall occupation crossing, wall-confined oppositely oriented signed response, and a finite mean damping gap. It does not establish a gapless Lindbladian pole, a quantized Hall coefficient, physical chirality of individual adaptive trajectories, or a wall-localized zero Lyapunov exponent. Those require record-resolved tangent or Choi transfer data, system-size scaling after removing exact null sectors, and a directed probe propagated through the same Born record. The present mean calculation is therefore a controlled baseline and an analytic counterexample to conflating stationary correlation-matrix topology with dynamical gaplessness.

VIII. REPRODUCTION CHECKLIST

The authoritative run root is the production directory named in Table I; its manifest status is **complete**. The note uses the archived analysis PDFs directly. Figures 1 and 3 are regenerated by `make_supplementary_figures.py`, which imports the pinned campaign solver, reconstructs $G(k_y) = C_{\text{arch}}^T(k_y)$ for shell one and the full frame, annotates the archived response, and verifies all 132 production checksums before and after plotting. It leaves all production files untouched.

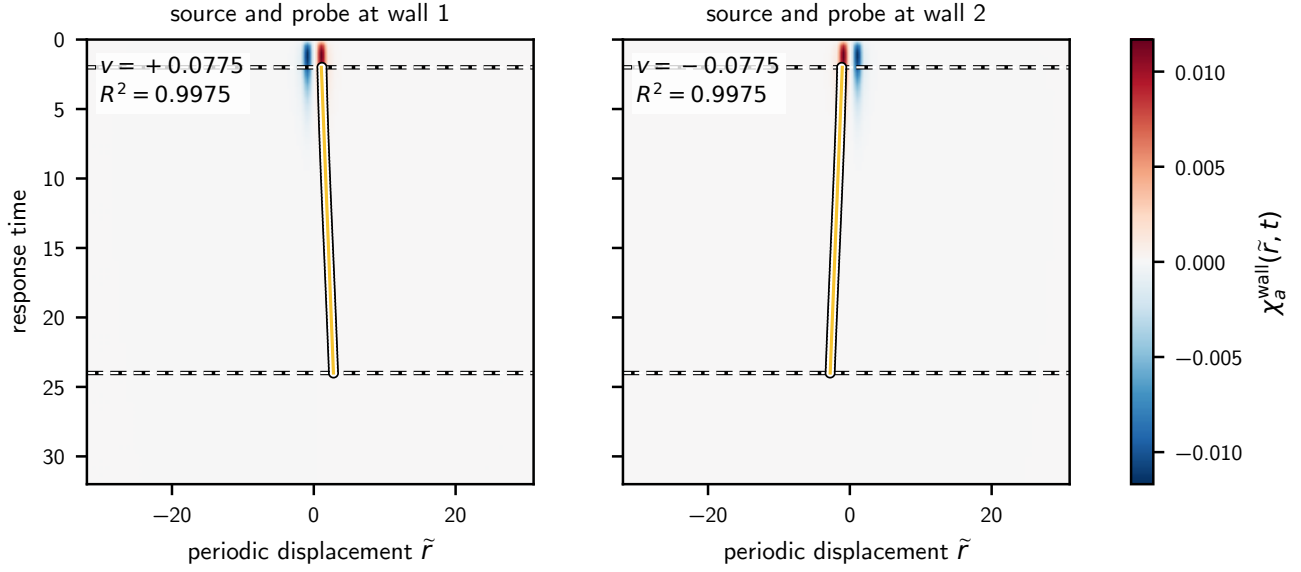


FIG. 3. Exact continuous-time response χ_a^{wall} for the canonical full-frame $N_y = 64$ endpoint. Each panel uses a separate kick placed at its named wall. White points show Eq. (51), yellow lines show the velocity fit, and horizontal dashed lines bound its time window. The opposite signed-dipole orientations coexist with exponential extinction.

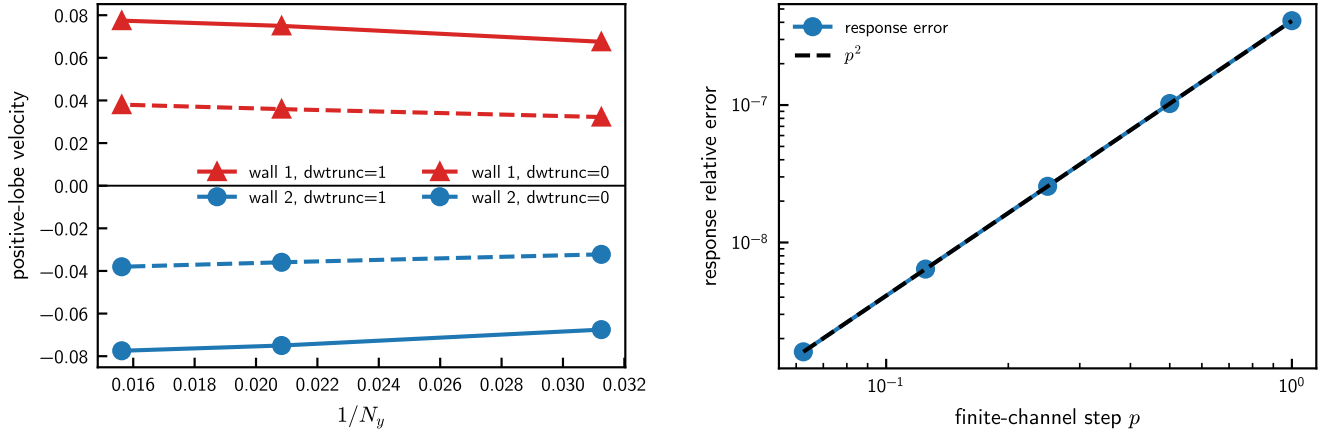


FIG. 4. Left: positive-lobe velocities for $N_y = 32, 48, 64$ in full-frame wall calculations, with hard domain-wall truncation (solid) and without it (dashed). The two wall orientations remain opposite; truncation strengthens the measured drift. These are raw size points, not an extrapolation. Right: $E_\chi(p)$ from Eq. (53); the dashed p^2 guide is anchored at the smallest saved p .

Compilation uses RevTeX 4.2 in double-column PRB for-

mat. The acceptance gate for this working note is a clean PDF containing fewer than ten pages.

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